Exercise 1.3

Miscellaneous Practice Problems

1. Every even number greater than 2 can be expressed as the sum of two prime numbers. Verify this statement for every even number upto 16.
2. Is 173 a prime? Why?
3. For which of the numbers, from n = 2 to 8, is $2n-1$ a prime?
4. State true or false and explain your answer with reason for the following statements.
   a) A number is divisible by 9, if it is divisible by 3.
   b) A number is divisible by 6, if it is divisible by 12.
5. Find A as required:
   (i) The greatest 2 digit number 9A is divisible by 2.
   (ii) The least number 567A is divisible by 3.
   (iii) The greatest 3 digit number 9A6 is divisible by 6.
   (iv) The number A08 is divisible by 4 and 9.
   (v) The number 225A85 is divisible by 11.
6. Numbers divisible by 4 and 6 are divisible by 24. Verify this statement and support your answer with an example.
7. The sum of any two successive odd numbers is always divisible by 4. Justify this statement with an example.
8. Find the length of the longest rope that can be used to measure exactly the ropes of length 1m 20cm, 3m 60cm and 4m.

Challenge Problems

9. The sum of three prime numbers is 80. The difference of two of them is 4. Find the numbers.
10. Find the sum of all the prime numbers between 10 and 20 and check whether that sum is divisible by all the single digit numbers.
11. Find the smallest number which is exactly divisible by all the numbers from 1 to 9.
12. The product of any three consecutive numbers is always divisible by 6. Justify this statement with an example.
13. Malarvizhi, Karthiga and Anjali are friends and natives of the same village. They work at different places. Malarvizhi comes to her home once in 5 days. Similarly, Karthiga and Anjali come to their homes once in 6 days and 10 days respectively. Assuming that they met each other on the 1st of October, when will all the three meet again?

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